Rational Exponents and Radical Graphs · Practice Quiz

Advanced Algebra · rational exponents, multistep expressions, radical graphs

Name: Date: Period:

Time: 30 minutes · 40 points. No calculator for Parts A and B. Show your work in Parts C, D and E: a correct answer with no work earns at most half the points. Assume all variables are positive. Write final answers with positive exponents unless the question asks for radical form.

Part A · Evaluate without a calculator (1 point each, 6 points)

Write "not real" if the value is not a real number.

1. $81^{1/4}$

2. $125^{2/3}$

3. $(-8)^{5/3}$

4. $16^{-3/2}$

5. $\left(\frac{1}{27}\right)^{-2/3}$

6. $(-81)^{1/4}$

Part B · Rewrite (1 point each, 4 points)

7. $\sqrt[5]{x^3}$ in rational-exponent form

8. $\left(\sqrt[3]{7}\right)^4$ in rational-exponent form

9. $y^{-2/3}$ in radical form

10. $10^{0.75}$ in radical form

Part C · Solve (2 points each, 8 points)

Give every real solution.

11. $x^3 = -216$ **(2 points)**

Work:

12. $3x^4 = 48$ **(2 points)**

Work:

13. $x^{3/2} = 27$ **(2 points)**

Work:

14. $2(x-1)^{2/3} = 18$ **(2 points)**

Work:

Part D · Simplify (2 points each, 12 points)

15. $5^{1/2} \cdot 5^{3/2}$ **(2 points)**

Work:

16. $\left(x^{2/3}\right)^{3/4} \cdot x^{1/2}$ **(2 points)**

Work:

17. $\left(16x^{8}y^{-4}\right)^{3/4}$ **(2 points)**

Work:

18. $\left(27a^{-6}b^{3}\right)^{-2/3}$ **(2 points)**

Work:

19. $\dfrac{\left(x^{1/2}y^{-1/3}\right)^{6}}{x^{2}y^{1/3}}$ **(2 points)**

Work:

20. $\sqrt[3]{54} + 2\sqrt[3]{16}$ **(2 points)**

Work:

Part E · Radical graphs (10 points)

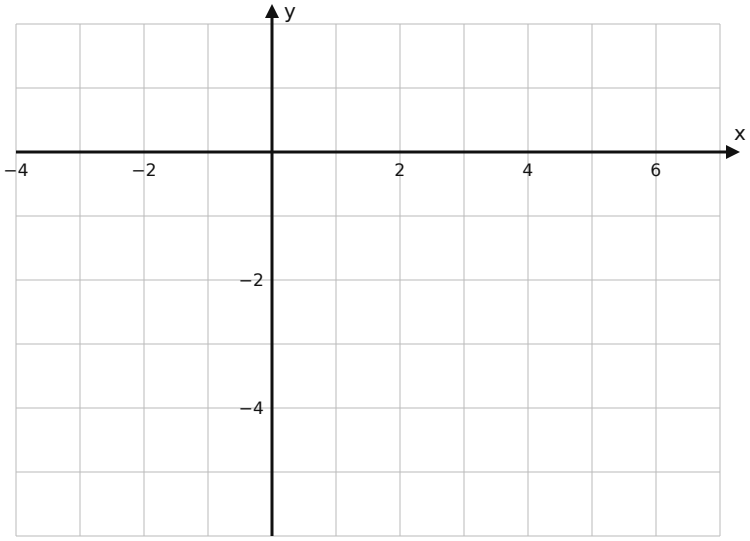
21. Let $g(x) = -2\sqrt{x+3} + 1$. **(6 points)**

(a) Describe the transformations from $f(x) = \sqrt{x}$. (2 points)

(b) Give the starting point, the domain and the range. (2 points)

(c) Complete the table and sketch the graph. (2 points)

x	−3	−2	1	6
$g(x)$				



22. Write a rule for g: the graph of $f(x) = \sqrt[3]{x}$ is stretched vertically by a factor of 3, then shifted 4 units right and 2 units down. **(2 points)**

23. Give the domain and range of (a) $h(x) = \sqrt[3]{x-2} + 5$ and (b) $k(x) = \sqrt{4-x}$. **(2 points)**